Analyzing Running Times
Of
Iterative Program Code

Find Big O

```
int i;
for ( i = 1 to n )
 print ("message")
```

Find Big O

0(n)

```
int i;
for ( i = 1 to n )
 print ("message")
```

```
int i,j;
for ( i = 1 to n )
for ( j = 1 to n )
 print (“message”)
```

$0(n^2)$

```
int i,j;
for ( i = 1 to n )
for ( j = 1 to n )
 print ("message")
```

```
fun(int n)
{
int i,j,k;
for ( i = 1 to n )
  {
    for ( j = 1 to i )
      {
           for(k=1; k<=100; k++)
            print ("message");
       }
    }
}
```

$0(n^2)$

```
fun(int n)
{
int i,j,k;
for ( i = 1 to n )
  {
    for ( j = 1 to i )
      {
          for(k=1; k<=100; k++)
           print ("message");
       }
     }
}
```

```
fun(int n )
{
 while (n >1)
     n= n/2
}
```

0(log n)

```
fun()
{
 while (n >1)
     n= n/2
}
```

```
fun(int n)
{
int i,j,k;
for ( i = 1 to n )
  {
    for ( j = 1; j<=i²: i++ )
      {
          for(k=1; k<=n/2; k++)
           print ("message");
       }
     }
}
```

$O(n^4)$

```
fun(int n)
{
int i,j,k;
for ( i = 1 to n )
  {
    for ( j = 1; j<=i²: i++ )
      {
          for(k=1; k<=n/2; k++)
           print ("message");
       }
     }
}
```

```
fun()
{
for ( i = 1 ; i<n ; i= i * 2 )
print ("message");
}
```

0(log n)

```
fun()
{
for ( i = 1 ; i<n ; i= i * 2 )
print (“message”);
}
```

```
fun(int n)
{
int i,j,k;
for ( i = n/2; i<=n ; i++ )
  {
    for ( j = 1; j<=n/2: j++ )
      {
           for(k=1; k<=n; k=k*2)
            print ("message");
       }
     }
}
```

$0(n^2 \log_2 n)$

```
fun(int n)
{
int i,j,k;
for ( i = n/2; i<=n ; i++ )
  {
    for ( j = 1; j<=n/2: j++ )
      {
          for(k=1; k<=n; k=k*2)
           print ("message");
       }
     }
}
```

```
fun(int n)
{
 i=1 ; sum=0;
 while (sum<=n)
        {
         sum=sum+i ;
          i++;
                 }
}
```

$0(n^{1/2})$

```
fun(int n)
{
 i=1 ; sum=0;
 while (sum<=n)
        {
         sum=sum+i ;
          i++;
                }
}
```

$0(n^{1/2})$

```
fun(int n)
{
 i=1 ;
 for (i=1; i²<=n, i++)
          {
            print ("message")
          }
}
```